BASE CHANGE OF THE UPPER SHRIEK ALONG A FLAT HOMOMORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-dualizing/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The double base change identity for the derived tensor product along the two polynomial presentations. Invoked inside the counted proof (source lines 4693-4695). Reduces the general statement to the case of the polynomial ring.
- The base change map for the upper shriek constructed in the trivial duality section. Invoked inside the counted proof (source lines 4696-4704). Supplies the map whose isomorphism is asserted.
- The base change statement for the upper shriek of a surjection, which makes that map an isomorphism on bounded below complexes. Invoked at the end of the counted proof (source lines 4705-4706). Upgrades the constructed map to an isomorphism in the bounded below case.

Source reference. Original equation equation-base-change: sources/stacks-dualizing/pinned-original-source.tex, source lines 2669–2669. Navigation pointer only: the counted text refers to the Stacks item equation-base-change, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-flat-bc-surjection: sources/stacks-dualizing/pinned-original-source.tex, source lines 2726–2726. Navigation pointer only: the counted text refers to the Stacks item lemma-flat-bc-surjection, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

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Source reference. Original reference more-algebra-lemma-double-base-change: sources/stacks-dualizing/pinned-original-source.tex, source lines 4695–4695. Navigation pointer only: the counted text refers to the Stacks item more-algebra-lemma-double-base-change, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original section section-base-change-trivial-duality: sources/stacks-dualizing/pinned-original-source.tex, source lines 2656–2656. Navigation pointer only: the counted text refers to the Stacks item section-base-change-trivial-duality, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $R \to R'$ be a flat homomorphism of Noetherian rings. Let $\varphi : R \to A$ be a finite type ring map. Let $\varphi' : R' \to A' = A \otimes_R R'$ be the map induced by φ. Then we have a functorial maps*

$$\varphi^!(K) \otimes^{\mathbf{L}}_A A' \longrightarrow (\varphi')^!(K \otimes^{\mathbf{L}}_R R')$$

for K in $D(R)$ which are isomorphisms for $K \in D^+(R)$.

Proof. Choose a factorization $R \to P \to A$ where P is a polynomial ring over R. This gives a corresponding factorization $R' \to P' \to A'$ by base change. Since we have $(K \otimes^{\mathbf{L}}_R P) \otimes^{\mathbf{L}}_P P' = (K \otimes^{\mathbf{L}}_R R') \otimes^{\mathbf{L}}_{R'} P'$ by More on Algebra, Lemma more-algebra-lemma-double-base-change it suffices to construct maps

$$R\operatorname{Hom}(A, K \otimes^{\mathbf{L}}_R P[n]) \otimes^{\mathbf{L}}_A A' \longrightarrow R\operatorname{Hom}(A', (K \otimes^{\mathbf{L}}_R P[n]) \otimes^{\mathbf{L}}_P P')$$

functorial in K. For this we use the map (equation-base-change) constructed in Section section-base-change-trivial-duality for P, A, P', A'. The map is an isomorphism for $K \in D^+(R)$ by Lemma lemma-flat-bc-surjection. □